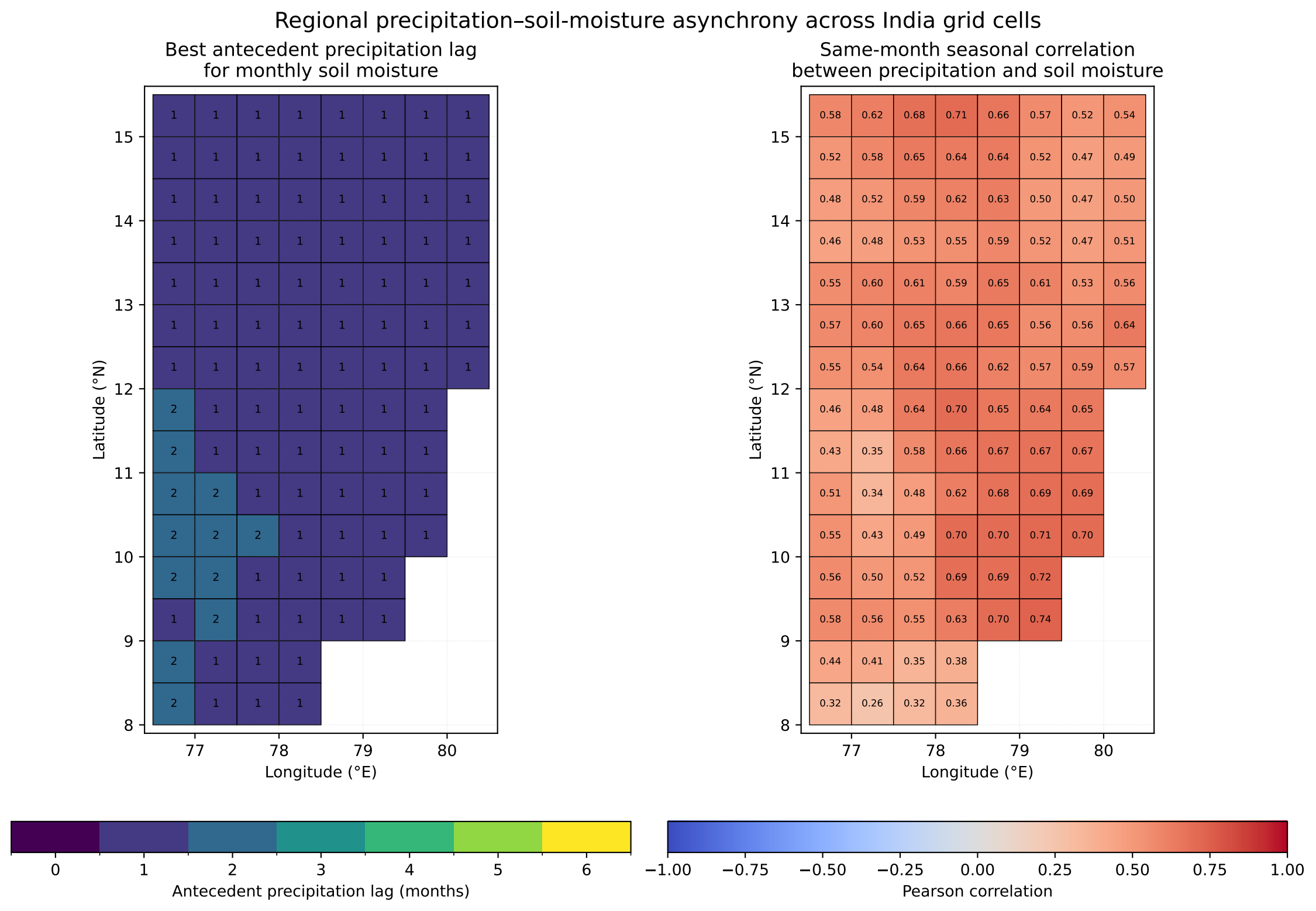
Regional precipitation–soil-moisture asynchrony across India grid cells
Best antecedent precipitation lag
for monthly soil moisture
Same-month seasonal correlation
between precipitation and soil moisture
Latitude (°N)
Longitude (°E)
15
14
13
12
11
10
9
8
77
78
79
80
1 1 1 1 1 1 1 1
1 1 1 1 1 1 1 1
1 1 1 1 1 1 1 1
1 1 1 1 1 1 1 1
1 1 1 1 1 1 1 1
1 1 1 1 1 1 1 1
2 1 1 1 1 1 1
2 1 1 1 1 1 1
2 2 1 1 1 1 1
2 2 2 1 1 1 1
2 2 1 1 1 1
1 2 1 1 1 1
2 1 1 1
2 1 1 1
0.58 0.62 0.68 0.71 0.66 0.57 0.52 0.54
0.52 0.58 0.65 0.64 0.64 0.52 0.47 0.49
0.48 0.52 0.59 0.62 0.63 0.50 0.47 0.50
0.46 0.48 0.53 0.55 0.59 0.52 0.47 0.51
0.55 0.60 0.61 0.59 0.65 0.61 0.53 0.56
0.57 0.60 0.65 0.66 0.65 0.56 0.56 0.64
0.55 0.54 0.64 0.66 0.62 0.57 0.59 0.57
0.46 0.48 0.64 0.70 0.65 0.64 0.65
0.43 0.35 0.58 0.66 0.67 0.67 0.67
0.51 0.34 0.48 0.62 0.68 0.69 0.69
0.55 0.43 0.49 0.70 0.70 0.71 0.70
0.56 0.50 0.52 0.69 0.69 0.72
0.58 0.56 0.55 0.63 0.70 0.74
0.44 0.41 0.35 0.38
0.32 0.26 0.32 0.36
0
1
2
3
4
5
6
Antecedent precipitation lag (months)
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
Pearson correlation